

ABERDEEN DYCE, United Kingdom

WMO: 030910

Lat: 57.2050N

Lon: 2.2053W

Elev: 66

StdP: 100.54

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	-2.4	-6.4	2.2	-2.3	-5.0	2.5	-0.4	15.5	6.2	14.2	5.8	1.2	0	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	21.9	16.7	20.1	15.5	18.8	14.7	17.5	20.8	16.4	19.1	15.6	17.9	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.5	18.9	15.2	10.9	17.8	14.4	10.3	16.8	49.4	20.9	46.3	19.2	43.8	18.0	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	10.9	9.7	-8.4	25.2	3.4	1.6	-10.8	26.4	-12.8	27.3	-14.7	28.2	-17.2	29.3
			-8.7	19.3	3.4	1.0	-11.2	20.0	-13.2	20.6	-15.1	21.2	-17.6	22.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	3.9	4.2	5.5	7.3	9.6	12.4	14.5	14.3	12.5	9.5	6.3	4.0
	DBStd	4.70	2.60	2.90	2.91	2.71	2.60	2.37	2.24	2.09	2.25	2.64	2.95	3.43
	HDD10.0	976	189	164	143	88	38	6	0	1	4	41	115	186
	HDD18.3	3521	447	396	398	331	269	178	119	124	176	274	362	444
	CDD10.0	499	0	1	3	8	27	77	141	135	78	25	3	1
	CDD18.3	4	0	0	0	0	0	0	2	1	1	0	0	0
	CDH23.3	17	0	0	0	0	0	2	9	6	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	4.5	5.1	5.2	5.1	4.6	4.2	4.2	3.8	3.9	4.3	4.6	4.6	4.8
Precipitation	PrecAvg	813	78	56	56	55	61	60	65	74	74	86	83	71
	PrecMax	1092	159	154	103	98	154	148	135	154	261	237	274	144
	PrecMin	516	12	8	10	14	13	18	12	12	20	9	21	18
	PrecStd	143	38	35	25	21	35	32	32	39	46	55	55	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	14.3	15.7	17.7	20.1	22.6	24.5	23.5	22.2	17.7	14.7	13.2
		MCWB	9.6	10.4	10.1	12.4	14.4	16.9	18.4	17.8	16.8	14.2	11.6	10.6
	2%	DB	10.6	12.0	13.3	15.2	17.6	19.8	22.2	21.0	19.2	15.6	12.8	11.6
		MCWB	8.5	9.1	9.2	10.7	12.9	14.6	16.9	16.5	15.0	12.6	10.6	9.5
	5%	DB	9.2	10.1	11.6	13.3	15.8	18.1	20.1	19.3	17.6	14.3	11.7	10.1
		MCWB	7.3	7.6	8.2	9.6	11.8	13.8	15.9	15.2	13.8	11.9	9.8	8.4
	10%	DB	7.9	8.6	10.2	11.9	14.2	16.6	18.6	18.1	16.2	13.3	10.6	8.7
		MCWB	6.3	6.3	7.3	8.6	10.6	12.8	14.9	14.5	13.1	11.3	9.2	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.8	11.1	13.5	15.3	17.4	19.2	18.6	17.8	14.7	12.2	11.0
		MCDB	11.7	13.5	14.3	17.0	19.0	21.8	23.6	22.8	21.0	17.0	13.6	12.9
	2%	WB	8.7	9.3	9.7	11.4	13.6	15.5	17.7	17.2	15.7	13.4	11.1	9.8
		MCDB	10.3	11.7	12.6	14.3	17.0	19.0	21.1	20.1	18.4	15.0	12.3	11.3
	5%	WB	7.6	8.0	8.5	10.1	12.2	14.5	16.6	16.2	14.6	12.6	10.2	8.6
		MCDB	8.8	9.9	11.1	12.8	15.1	17.3	19.2	18.4	16.6	13.8	11.2	9.8
	10%	WB	6.5	6.7	7.5	9.0	11.1	13.4	15.6	15.3	13.8	11.8	9.4	7.5
		MCDB	7.7	8.3	9.9	11.4	13.6	16.0	18.0	17.2	15.5	12.9	10.4	8.7
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.5	6.2	6.6	7.1	6.5	6.7	6.8	6.5	5.8	5.4	5.1
		MCDBR	6.5	7.4	8.9	9.0	9.5	9.2	9.5	8.7	8.7	7.1	6.5	6.3
		MCWBR	5.7	5.6	5.9	5.7	5.8	5.6	5.6	5.2	5.5	5.3	5.6	5.6
	5% WB	MCDBR	6.1	7.0	8.4	8.6	8.9	8.6	8.4	8.0	7.6	6.4	6.0	6.0
		MCWBR	5.5	5.6	6.0	5.8	5.8	5.7	5.4	5.2	5.3	5.2	5.4	5.5
Clear-Sky Solar Irradiance	taub	0.306	0.331	0.357	0.379	0.375	0.373	0.377	0.366	0.350	0.337	0.310	0.290	
	taud	2.265	2.350	2.338	2.303	2.368	2.398	2.392	2.431	2.481	2.437	2.309	2.201	
	Ebn at Noon	559	716	794	832	862	867	854	840	801	703	550	459	
	Edh at Noon	51	71	92	111	110	109	107	97	80	64	48	40	
All-Sky Solar Radiation	RadAvg	0.44	1.08	2.17	3.59	4.85	4.77	4.63	3.68	2.54	1.33	0.58	0.31	
	RadStd	0.06	0.10	0.24	0.34	0.26	0.39	0.36	0.24	0.20	0.12	0.05	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page